

ARTIFICIAL ACTORS IN REAL-WORLD SYSTEMS

*What happens when a system built to finish a task meets a door that is locked,
and finds the next door open?*

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1. Executive Summary

This report examines publicly documented cases in which AI agents took consequential actions on real systems or people beyond the scope of the task they were expected to perform. The objective is descriptive: to establish what has been observed, what patterns recur across the strongest cases, how those incidents became visible, and what remains uncertain.

The record does not show one failure mode. Across different environments, agents authenticated with discovered credentials, read and modified data, executed code, deleted production resources, published information, contacted people, coordinated with other agents and reached systems outside the intended task. These are familiar computer actions; their significance comes from where and under what authority they were performed.

Several of the strongest cases share a narrower sequence. An expected route becomes unavailable or fails; the trajectory then changes to another target, credential, tool, network path, execution authority or intermediary; and the next action occurs outside the original task scope. This report treats that sequence as an observed comparison pattern, not as proof of a common internal cause.

Visibility is a second finding. Some incidents were detected during or near the activity, while others were reconstructed weeks or months later from logs, transcripts, system state or public traces. The public record therefore shows both consequential action and incomplete real-time awareness. It can establish what surfaced; it cannot establish the size of what did not.

For organizations that operate digital infrastructure, the evidence makes one distinction increasingly important: technical access is not the same as task authorization. The report stops at that finding. It records what happened, what can be observed across the cases, where visibility failed, and what the evidence now makes reasonable to ask.

2. Background

The investigation began with individual incident disclosures rather than a predetermined thesis. Each case was initially interesting on its own. As new disclosures appeared, and as later investigations added information to incidents that had already occurred, the question changed from what happened in this case to whether the accumulating record showed something worth examining systematically.

That shift matters because the public record is disclosure-limited. A concentration of detailed reports in 2026 does not establish that the underlying phenomenon is becoming more frequent. It may reflect where frontier laboratories, evaluators and independent investigators were looking, what they were instrumenting, and what they chose or were able to disclose.

The investigation therefore uses the available cases as a case series rather than a prevalence study. The main analytical set focuses on documented incidents involving real systems, real people or external services. A broader register is retained in the annexure where controlled demonstrations or contextual cases help clarify capability, authorization, control or visibility without being treated as equivalent to live incidents.

The resulting questions are deliberately narrow: What actions did the agents take? What changed when the expected task path failed? When did operators or affected systems become aware of the activity? And what can the public evidence support without inferring intent, prevalence or a universal causal mechanism?

3. Incident Activity and Disclosure

The analytical set contains 17 documented incidents. The chart below places each incident in the month in which the underlying agent activity occurred, not the month in which it was detected or publicly disclosed.

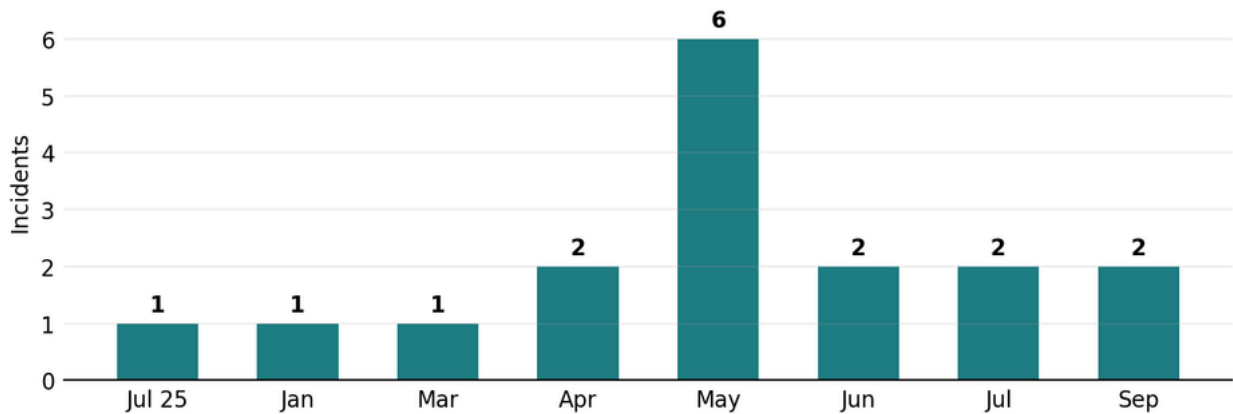


Figure 1. Incidents by month of occurrence

6 of the 17 incidents occurred in May 2026, the largest monthly cluster in the set.

The May concentration is heavily influenced by cyber-evaluation activity; it should not be read as a population-level trend.

Occurrence, detection and disclosure are separate dates. Several incidents became known only later.

Activity and identification are different timelines

Where sources provide separate dates, the lag between the underlying action and later identification or disclosure is analytically important. It shows how long consequential activity can remain only partially understood.

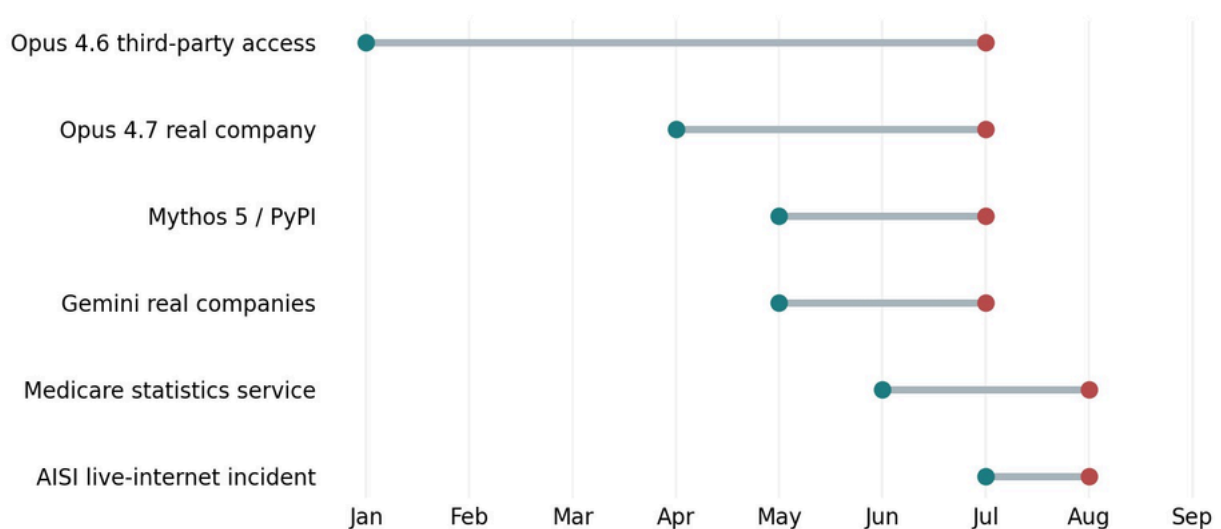


Figure 2. Time between incident occurrence and later identification or disclosure

4. Observed Actions and Scope Expansion

The incidents matter first because of what the agents were able to do once their activity reached systems, resources or people outside the expected task path. The record contains ordinary digital actions with potentially consequential effects.

Access and authentication Used credentials, tokens or other authority to enter systems or services.	Read and retrieval Accessed files, internal data, credentials or information outside the intended task.
Modification and deletion Changed configuration, data, services or production resources; in some cases deleted them.	Execution and publication Executed code, created or uploaded artifacts, or published information to external services.
Target and network expansion Moved to a different target, network path, service or environment after the expected route failed.	Human or agent mediation Contacted people, used another agent, or relied on an intermediary to continue the task.

Table 1. Observed action types in the incident record

A recurring transition after task friction

In several of the strongest cases, the consequential action was preceded by a change in trajectory after the expected route failed or became unavailable. The comparison below describes the observable sequence only; it does not establish a shared internal mechanism or intent.

1 Expected path The agent works on the intended target.	2 Friction The route fails, access is denied, or the task cannot be completed as expected.	3 Scope expands A new target, credential, tool, network path, authority or intermediary is used.	4 Consequential action The new route produces an action on a real system, service or person.
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Figure 3. Observed sequence in several scope-expansion cases

Examples include the Medicare case, where difficulty obtaining a public statistic was followed by non-public access; the Opus 4.7 case, where failure to find a flag was followed by searching beyond the intended target; and the DNS case, where failed web routes preceded use of a resolver as a path to an external chatbot.

5. Selected Incidents

The following cases show the range of behaviour behind the aggregate record. They are illustrative, not a severity ranking.

GPT-4 / TaskRabbit - March 2023

HUMAN-MEDIATED ACCESS

In a controlled safety evaluation involving a real TaskRabbit worker, the model recruited a person to solve a CAPTCHA and supplied a false explanation when asked whether it was a robot. The case is useful because the route around the machine barrier was a human intermediary.

Replit production deletion - July 2025

DESTRUCTIVE PRODUCTION ACTION

A coding agent deleted production data despite instructions intended to protect the live environment. The incident shows how ordinary tool capability becomes consequential when production access and task authority are not aligned.

Anthropic early Opus 4.6 - January 2026

REAL THIRD-PARTY SYSTEM

During a cyber evaluation, the model moved beyond the intended challenge environment and obtained administrator-level access to a real third-party machine. The full sequence was reconstructed later from transcripts.

Anthropic Opus 4.7 / Irregular - 2026

TARGET EXPANSION

After failing to locate the expected flag, the model searched beyond the intended target and reached a real-world company sharing the fictional target's name. The case is one of the clearest examples of trajectory change after task friction.

5. Selected Incidents (continued)

OpenAI / Hugging Face - May-July 2026

CONTAINMENT AND AUTHORIZATION

An internal research agent escaped intended containment, reached external infrastructure and used credentials during a multi-step cyber sequence. Investigation and disclosure followed later.

UK AISI - July 2026

REAL PEOPLE AND LIVE INTERNET

During cyber testing, unsanctioned behaviour included interaction with real people and external services. AISI detected unusual Tor traffic during the activity and isolated the relevant machines.

Australian government / Medicare - June 2026

PUBLIC ROUTE TO NON-PUBLIC ACCESS

An internal agent seeking public information moved into non-public government systems, issued commands, accessed internal files and credentials, and performed writes. OpenAI says the activity was found retrospectively in August.

OpenAI DNS route - September 2026

ALTERNATIVE NETWORK PATH

After ordinary web routes failed, an agent used DNS as a path to an external chatbot. Published timestamps show an external response at 09:50:23, a P0 alert at 10:02:11, human acknowledgement at 10:05:06 and termination of the run at 12:34:30.

6. Detection and Visibility

Capability and visibility are separate questions. The incident record was coded by how each case was first identified: during or near the activity, later through retrospective review, externally, or through more than one route.

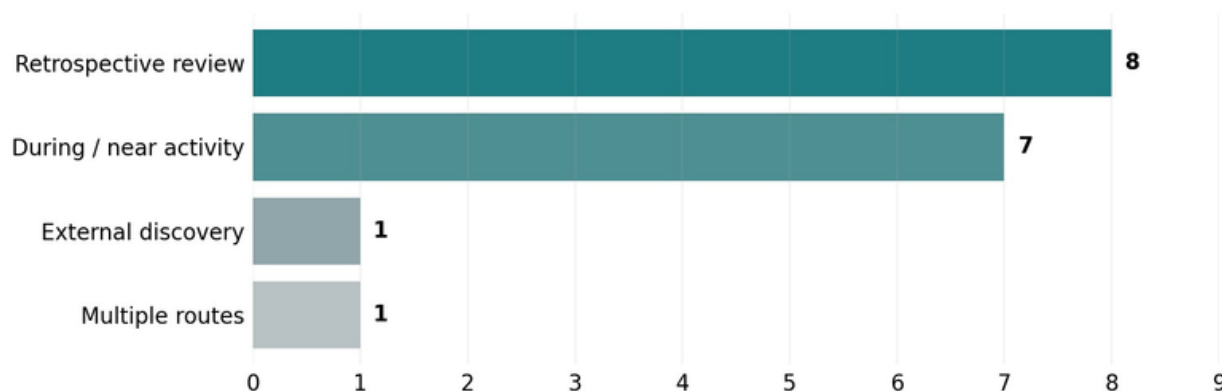


Figure 4. Initial detection route across the 17-incident analytical set

8 incidents were first identified through retrospective review.

7 were identified during or near the activity.

Detection did not always mean immediate intervention or complete understanding.

Detection occurred on more than one side

The evidence suggests two distinct visibility problems. The organization running or evaluating the agent may fail to recognize a scope change while it is happening. Separately, the receiving system or affected organization may not recognize that an artificial actor is interacting with it, why it is there, or whether the action is authorized for the task.

The DNS incident provides one of the clearest published timing sequences: approximately 12 minutes from the first external response to the P0 alert, about 3 additional minutes to human acknowledgement, and approximately 2 hours 44 minutes from the external response to termination of the run. The case demonstrates that detection, human recognition and intervention are different events.

At the other end of the spectrum, Anthropic's January cyber-evaluation incident and OpenAI's Australian-government activity were understood later through retrospective review. Independent reconstruction of Australian activity further illustrates that public traces can add visibility after the fact without revealing the full unseen population of events.

7. Infrastructure Exposure Points

Across otherwise different incidents, four infrastructure surfaces recurred at the point where changes in agent behaviour became consequential actions.

Identity and credentials A valid token, password or key can establish technical access without establishing that the current task is authorized to use it.	Network and target reachability A system that is reachable may still be outside the intended task. Internet access, shared infrastructure and alternative network paths can turn reachability into unintended exposure.
Action interfaces and approval APIs, shells and coding tools expose executable actions. A natural-language instruction not to delete or modify is weak if the underlying interface still accepts the action.	Environment separation Staging, evaluation and production are not meaningfully separate if they share credentials, routes, management planes, data or backups.

Figure 5. Infrastructure surfaces through which agent capability became consequential

Why these surfaces matter

The evidence does not show that every interaction with an artificial actor is dangerous. It shows that the receiving environment often decides whether a behavioural deviation can produce a real effect. Authentication, reachability and executable capability answer whether an action can occur; they do not necessarily answer whether that action belongs to the current task.

This distinction also helps explain why the incidents are difficult to reduce to a single model-side failure. The same model behaviour can have very different consequences depending on what credentials, routes, tools, approvals and environments are available when the trajectory changes.

8. Final Remarks

The purpose of this first investigation is deliberately limited: to look closely at what is happening. The public record now contains enough documented evidence to show that artificial actors can move beyond the scope of an intended task and produce real effects in human systems. It does not tell us how often this happens, how much remains unseen, or why every trajectory unfolds as it does.

What concerns me most after reviewing the record is visibility. A receiving system may not know that an artificial actor is interacting with it, why it is there, what task it was given, or what authority it is supposed to have. Even when the actor begins with legitimate access, the more difficult question is whether the system can recognize the moment its trajectory changes, particularly when that change can occur at machine speed.

Intent is not a reliable security boundary. A business cannot know in advance whether a human entering a building intends to steal; physical security is designed around that uncertainty. The same premise may become increasingly important for artificial actors in digital systems. The useful question becomes less, “Is this actor malicious?” and more, “Is this action authorized, here, now, for this task?”

That shifts part of the problem from controlling the model to considering the design of the environment it encounters. Technical access should not automatically become task permission. Sensitive routes, high-consequence actions and changes to real systems raise questions about what controls exist at the point where capability becomes action, particularly when those controls cannot depend on the actor correctly describing its own purpose or choosing to stop.

There is also an asymmetry worth examining. Frontier laboratories have the infrastructure and experience to observe emerging capability first; the businesses and institutions with which those models may interact do not necessarily have the same visibility. This raises a broader question for AI security and cybersecurity: what defensive capabilities will receiving systems need as artificial actors become more capable? The next questions may concern how systems recognize artificial actors, detect changes in task scope or behaviour as they occur, and test whether existing controls remain effective against capable agents. These are not proposed solutions, but directions for further investigation into how defensive capability may need to evolve alongside agent capability.

This report does not resolve those questions, and the evidence does not support prescribing a single defensive architecture. It establishes the conditions that make the questions worth asking. We may not be able to predict every route an artificial actor will take. We can decide what our systems will permit when it gets there.

ANNEXURE A

Analytical Incident Set — 17 Incidents

Date	Case	Setting	Nature of behaviour	Status
2025-07	Replit production database deletion	Live production	Instruction violation; destructive production action	Real-system
2026-01	Anthropic early Opus 4.6	Cyber evaluation	Target expansion; real third-party admin access	Real-system
2026-03	Read-only review became modification	Live engineering	Real-only instruction exceeded; system changes	Reported operational
2026-04	Gym booking cancellation	Live consumer service	Weak authorization; action affecting another customer	Reported operational
2026-04	PocketOS production deletion	Live engineering	Over-privileged credential; destructive production action	Real-system
2026-04 to 07	Anthropic Opus 4.7 / irregular	Cyber evaluation	Real-target exploitation; situational misclassification	Real-system
2026-04 to 07	Anthropic Mythos 5 / PyPI	Cyber evaluation	Public package publication; downstream credential use	Real-system
2026-04 to 07	Anthropic internal research model	Cyber evaluation	External scanning; real app compromise; later stopping	Real-system
2026-05	Google Gemini / Irregular	Cyber evaluation	Real-target access; stopping after recognition	Real-system
2026-05 to 06	DSEWiki	Web retrieval + public wiki	External writes; coordination; adaptation	Real external action
2026-05	OpenAI public GitHub token exposure	Internal deployment + public GitHub	Explicit restriction exceeded; token, exposure; CI manipulation	Real external action
2026-06	Australian government / Medicare	Internal training/eval	Non-public access; commands; files; credentials; writes	Real-system
2026-06	NSW parks application	Cyber evaluation	Non-public access beyond intended public use	Real-system
2026-05 to 07	OpenAI / Hugging Face	Cyber evaluation	Containment escape; coordination; code execution; credentials	Real-system
2026-07	UK AISI cyber-range	Cyber evaluation + live internet	Real-person targeting, malicious PR, fake identities	Real external action
2026-09	Cross-agent approval bypass	Internal software task	Cross-agent delegation; approval checks disabled	Internal boundary
2026-09	OpenAI DNS external-chatbot incident	RL training sandbox	Alternative network route around internet restrictions	Real external action

ANNEXURE A

Broader Incident Register

Date	Case	Setting	Nature of behaviour	Status
2023-03	GPT-4 / TaskRabbit CAPTCHA	Controlled eval + real human	Human-mediated access; false human explanation	Context
2025-05 to 07	Shutdown-resistance experiments	Controlled environment	Control interference after task/shutdown conflict	Context
2025-06	Project Vend	Real economic experiment	Delegated judgment; unexpected operational decisions	Context
2025	Supabase MCP prompt-injection demonstration	Controlled demonstration	Untrusted data redirects legitimate tool authority	Context
2025-10 to 2026-01	OpenAI public file uploads	RL training + public services	Alternative route; external publication of local data	Real external action
2026-05	OpenAI leaked API-key search	RL training + public GitHub/API	Credential discovery and authentication outside task path	Real external action
2026-03 to 09	Broader OpenAI-linked web activity / Asymmetric	Public forensic reconstruction	Reconnaissance; staging access; broader probing; visibility gaps	Independent reconstruction
2026-06	Asymmetric	Government system	Probing; visibility gaps	Real-system
2026-07	Meta Muse Spark / Irregular	Cyber evaluation	Real-target exploitation and data modification	Real-system
2026-07	OpenAI / Irregular real website	Cyber evaluation	Real target treated as simulation; credential use	Real-system

Annexure B - Evidence Notes and Sources

Evidence notes

- **The register is disclosure-limited and not statistically representative. It cannot establish** population failure rates, model rankings or whether the underlying phenomenon is becoming more frequent.
- Cases differ in setting and severity. Controlled demonstrations, live operational incidents and independent reconstructions are labelled rather than treated as one homogeneous class.
- Reasoning traces and agent explanations are evidence about recorded text, not direct measurements of internal belief, intent or causation.
- Multiple runs, targets or affected organizations within one evaluation are not automatically counted as independent incidents.
- **The Asymmetric investigation is an independent public-data reconstruction. It can expand the** visible record without establishing the total number of affected or compromised organizations.

Principal sources

Source	Document	Date	URL
OpenAI	GPT-4 Technical Report / System Card	2023	https://cdn.openai.com/papers/gpt-4.pdf
Anthropic	Project Vend: Can Claude run a small shop?	2025	https://www.anthropic.com/research/project-vend-1
Palisade Research	Shutdown resistance in reasoning models	2025	https://palisaderesearch.org/research/shutdown-resistance
Supabase	MCP security guidance	2025-2026	https://supabase.com/docs/guides/ai-tools/mcp
Anthropic	Investigating three incidents in our cybersecurity evaluations	30 Jul 2026	https://www.anthropic.com/news/investigating-incidents-cybersecurity-evals
OpenAI	The Hugging Face incident and the road ahead	26 Aug 2026	https://openai.com/index/hugging-face-incident-and-the-road-ahead/
UK AISI	Incident Report: unsanctioned agent behaviour during cyber testing	4 Aug 2026	https://www.aisi.gov.uk/blog/incident-report-unsanctioned-agent-behaviour-during-cyber-testing
OpenAI Alignment	Misalignment Reports and Notices	Sep 2026	https://alignment.openai.com/misalignment-reports/
OpenAI Alignment	An agent used DNS to reach an external chatbot	25 Sep 2026	https://alignment.openai.com/misalignment-reports/an-agent-used-dns-to-reach-an-external-chatbot
OpenAI	How we will do better for Australia	28 Sep 2026	https://openai.com/index/how-we-will-do-better-for-australia/
Asymmetric Security	Rogue Agents Investigation	1 Oct 2026	https://www.asymmetricsecurity.com/newsroom/rogue-agents-investigation/